

Name: Y.Veerendar Reddy

Reg.No:- 192325102

Subject Code/Name: MLA0305-Reinforcement Learning

EXPERIMENT – 2

Implementation of Reinforcement Learning Framework and Markov Decision Process (MDP)

Aim

To implement a basic Markov Decision Process (MDP) and demonstrate the fundamental components of a Reinforcement Learning framework, including states, actions, rewards, transitions, and policies.

Procedure

1. Define the states and available actions of the environment.
2. Define the transition probabilities between states.
3. Assign rewards for different state transitions.
4. Initialize the MDP environment.
5. Define a simple policy for selecting actions.
6. Execute the policy for several steps.
7. Record the states, actions, and rewards obtained.
8. Calculate the total accumulated reward.
9. Visualize the state transitions and rewards.

Code:-

```
# =====  
# EXPERIMENT 2  
# Implementation of Reinforcement Learning Framework  
# and Markov Decision Process (MDP)  
# =====  
  
import numpy as np
```

```
import matplotlib.pyplot as plt

# -----
# STEP 1: Define States
# -----

states = ["Start", "Middle", "Goal"]

# Assign numerical IDs to states
state_id = {
    "Start": 0,
    "Middle": 1,
    "Goal": 2
}

# -----
# STEP 2: Define Actions
# -----

actions = ["Move", "Stay"]

# -----
# STEP 3: Define Rewards
# -----

# Reward obtained for each state-action pair
rewards = {
    ("Start", "Move"): 1,
    ("Start", "Stay"): -1,
    ("Middle", "Move"): 10,
    ("Middle", "Stay"): -1,
```

```
    ("Goal", "Move"): 0,  
    ("Goal", "Stay"): 0  
}
```

```
# -----
```

```
# STEP 4: Define Transition Function
```

```
# -----
```

```
def transition(state, action):
```

```
    if state == "Start":  
        if action == "Move":  
            return "Middle"  
        else:  
            return "Start"
```

```
    elif state == "Middle":  
        if action == "Move":  
            return "Goal"  
        else:  
            return "Middle"
```

```
    elif state == "Goal":  
        return "Goal"
```

```
# -----
```

```
# STEP 5: Define a Simple Policy
```

```
# -----
```

```
def policy(state):
```

```
# Move until the goal is reached
if state == "Start":
    return "Move"

elif state == "Middle":
    return "Move"

else:
    return "Stay"

# -----
# STEP 6: Run the MDP
# -----

current_state = "Start"

state_history = []
action_history = []
reward_history = []

total_reward = 0

print("MDP Execution")
print("-" * 50)

for step in range(5):

    # Select action according to policy
    action = policy(current_state)
```

```

# Obtain reward
reward = rewards[(current_state, action)]

# Find next state
next_state = transition(current_state, action)

# Store information
state_history.append(current_state)
action_history.append(action)
reward_history.append(reward)

total_reward += reward

print(
    f"Step {step + 1}: "
    f"State = {current_state}, "
    f"Action = {action}, "
    f"Reward = {reward}, "
    f"Next State = {next_state}"
)

current_state = next_state

```

```

# -----
# STEP 7: Display Final Result
# -----

print("\n" + "=" * 50)
print("MDP RESULTS")

```

```

print("=" * 50)

print("State History :", state_history)
print("Action History:", action_history)
print("Reward History:", reward_history)
print("Total Reward :", total_reward)

# -----
# STEP 8: Visualization
# -----

plt.figure(figsize=(10, 5))

# Plot rewards obtained at each step
plt.plot(
    range(1, len(reward_history) + 1),
    reward_history,
    marker="o",
    linewidth=2
)

plt.title("Rewards Obtained During MDP Execution")
plt.xlabel("Step")
plt.ylabel("Reward")
plt.xticks(range(1, len(reward_history) + 1))
plt.grid(True)

plt.show()

# -----

```

```

# STEP 9: State Transition Visualization
# -----

plt.figure(figsize=(10, 5))

state_values = [state_id[state] for state in state_history]

plt.plot(
    range(1, len(state_values) + 1),
    state_values,
    marker="o",
    linewidth=2
)

plt.title("State Transitions in MDP")
plt.xlabel("Step")
plt.ylabel("State")

plt.yticks(
    [0, 1, 2],
    ["Start", "Middle", "Goal"]
)

plt.xticks(range(1, len(state_values) + 1))
plt.grid(True)

plt.show()

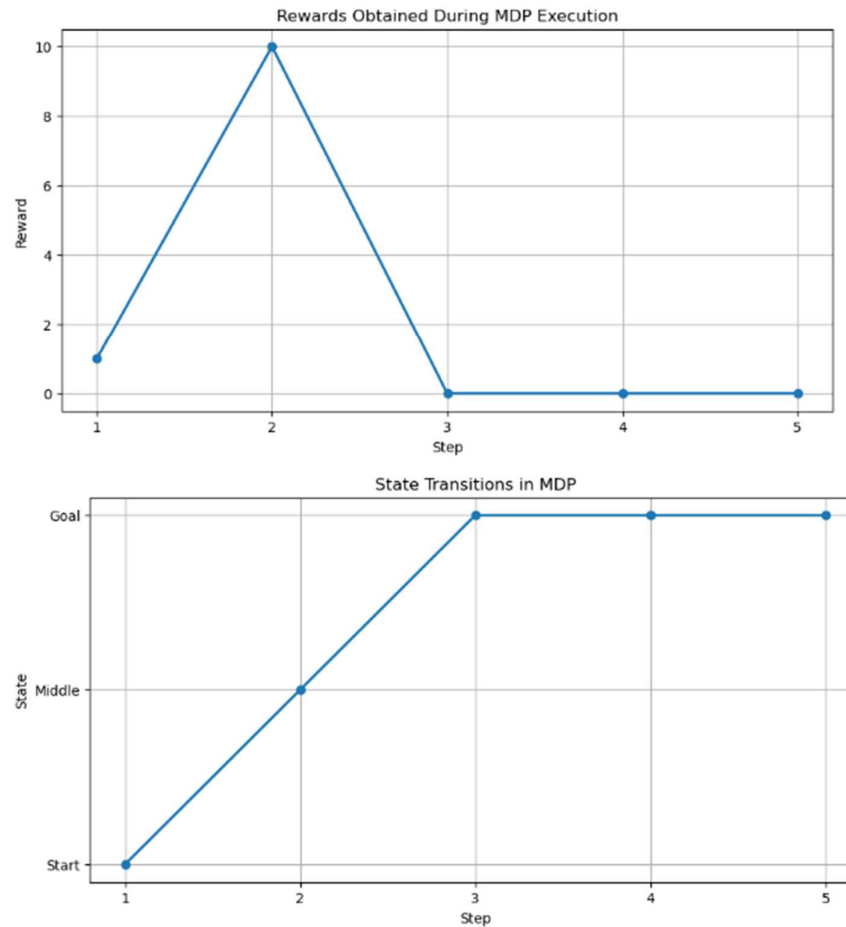
```

Output:-

```

MDP Execution
-----
Step 1: State = Start, Action = Move, Reward = 1, Next State = Middle
Step 2: State = Middle, Action = Move, Reward = 10, Next State = Goal
Step 3: State = Goal, Action = Stay, Reward = 0, Next State = Goal
Step 4: State = Goal, Action = Stay, Reward = 0, Next State = Goal
Step 5: State = Goal, Action = Stay, Reward = 0, Next State = Goal
=====
MDP RESULTS
-----
State History : ['Start', 'Middle', 'Goal', 'Goal', 'Goal']
Action History: ['Move', 'Move', 'Stay', 'Stay', 'Stay']
Reward History: [1, 10, 0, 0, 0]
Total Reward : 11

```



Visualization

The program produces two graphs:

1. Rewards Obtained During MDP Execution
 - X-axis → Steps
 - Y-axis → Reward
 - Shows the reward received at each decision step.
2. State Transitions in MDP
 - X-axis → Steps

- Y-axis → State
- Shows how the agent moves from Start → Middle → Goal.

Summary

The MDP framework was successfully implemented by defining states, actions, rewards, transition rules, and a policy. The agent followed the policy and moved toward the goal while accumulating rewards.

Result

Thus, the basic Reinforcement Learning framework and Markov Decision Process (MDP) were successfully implemented, and the state transitions and rewards were visualized.